🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 · STANDARD 6.NOS.4

Prime Factorization Flagship

Lesson 1-1-flagship · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write a number as a product of its prime factors using a factor tree.
Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Prime number | **Composite number** | **Prime factorization** | **Factor tree** | **Exponent**

👆 *Tap any word to see what it means and a picture.*

1. A whole number greater than 1 with exactly two factors, 1 and itself, is a ________ number.

2. A whole number greater than 1 that has more than two factors is a ________ number.

3. ________ — Writing a number as prime numbers multiplied together.

4. I can break a number into its prime factors step by step using a ________.

5. In 2^3, the small number 3 is the ________ and it shows 2 is multiplied 3 times.

Reflect — Exit Ticket

What is the prime factorization of 40?

A. 2 × 2 × 2 × 5

B. 4 × 10

C. 5 × 8

D. 2 × 20

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor tree
5. **Notes 5:** Exponent
6. **Watch for this mistake:** *A common mistake in Prime Factorization is skipping the key idea: "Keep breaking a number apart until every factor is a prime number you cannot split anymore." — always check your work against this rule before you submit.*
7. **Try It 1:** A. $2^3 \times 3^2$ — *$72 = 2 \times 2 \times 2 \times 3 \times 3$. Three 2s and two 3s give $2^3 \times 3^2$.*
8. **Try It 2:** A. $2 \times 2 \times 3 \times 7$ — *$84 = 4 \times 21 = (2 \times 2) \times (3 \times 7) = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$. The vault opens!*
9. **Exit Ticket:** A. $2 \times 2 \times 2 \times 5$ — *$40 = 2 \times 20 = 2 \times 2 \times 10 = 2 \times 2 \times 2 \times 5$. All factors (2, 2, 2, 5) are prime.*